$f : \mathbb{R}^n \longrightarrow \mathbb{R}$

$F : \mathbb{R}^n \longrightarrow \mathbb{R}^n$

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}(x), \ldots\ldots \frac{\partial f}{\partial x_n}(x) \right)$$ gradient

$$\nabla \cdot F(x) = \frac{\partial F_1}{\partial x_1}(x) + \ldots + \frac{\partial F_n}{\partial x_n}(x)$$ divergence

$$\nabla \times F(x) = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}, \frac{F_2}{\partial x} - \frac{F_1}{\partial y} \right)$$

$n = 3$

$$\Delta f = \frac{\partial^2 f}{\partial x_1^2} + \ldots + \frac{\partial^2 f}{\partial x_n^2}$$

$n = 3$

$\nabla \times \nabla f = 0$

$$\nabla \times \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = \left(\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2}{\partial z \partial y}, \dots, \dots \right) = \vec{0}$$

$$\nabla \cdot (\nabla \times F) = 0$$

$$\nabla \cdot (-y_z + z_y, -z_x + x_z, -x_y + y_x) =$$

$$\cancel{z_{yx}} - \cancel{y_{zx}} + \cancel{x_{zy}} - \cancel{z_{xy}} + \cancel{y_{xz}} - \cancel{x_{yz}} = 0$$

cotomological complex, ke imaš duc

zapoledn: pušči je 0

$$\mathbb{R} \xrightarrow{\nabla} \mathbb{R}^3 \xrightarrow{\nabla \times} \mathbb{R}^3 \xrightarrow{\nabla \cdot} \mathbb{R}$$

$K = [0,1] \times [0,1] \subseteq \mathbb{R}^2$

$$\int_K \nabla \cdot \vec{F}\, dx\, dy = \int_{\partial K} \vec{F}\, d\vec{n}$$

normale

$$\int_0^1 \int_0^1 \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} \right) dx\, dy = \int_0^1 dy \int_0^1 \frac{\partial F_1(x,y)}{\partial x} dx + \int_0^1 dx \int_0^1 \frac{\partial F_2(x,y)}{\partial y} dy$$

$$= \int_0^1 F_1(x,y)\Big|_0^1 dy + \int_0^1 F_2(x,y)\Big|_0^1 dx =$$

$$= \int_0^1 \left(F_1(1,y) - F(0,y) \right) dy + \int_0^1 \left(F_2(x,1) - F_2(x,0) \right) dx$$

$$\int_{\partial K} \vec{F}\, d\vec{n} = \int_0^1 \left(F_1(x,0), F_2(x,0) \right) \begin{pmatrix} dx \\ 0, -1 \end{pmatrix} dy + \dots$$

$$= \int_0^1 -F_2(x,0)\, dx + \dots = \int_0^1 \left(F_1(1,y) - F_1(0,y) \right) dy + \int_0^1 \left(F_2(x,1) - F_2(x,0) \right) dx$$

$u_{xy} = x^2 y \qquad u(x,0) = x \qquad u(0,y) = y^2$

$$u_x = \frac{1}{2}x^2y^2 + C(x)$$

$$u = \frac{1}{6}x^3y^2 + C(x) + D(y)$$

$$u(x,0) = C(x) + D(0) = x$$

$$u(0,y) = C(0) + D(y) = y^2$$

$$C(x) + D(y) = x + y^2$$

$C(0)$ in $D(0)$ inhlo dano za 0

$$u(x,y) = \frac{1}{6}x^3y^2 + x + y^2$$

$$u_{xy} + u_x = y e^{-y}$$

$$u(0,y) = 0$$

$$u(x,x) = 0$$

$$u_y(0,y) = 0$$

$$\frac{u(x,x)}{\partial y} = 0$$

integriramo po x

$$u_y + u = xye^{-y} + C(y)$$

v $(0,y) \rightsquigarrow$

$$0 + 0 = 0 + C(y) = 0$$

$$u_y + u = xye^{-y}$$

linearna diferencialna enačba

homogen del

$$u + u_y = 0$$

$$u = -\frac{\partial u}{\partial y} \quad dy = -\frac{du}{u} \rightsquigarrow y = -\ln u + C(x)$$

$$u = Ce^{-y}$$

partikularna

$$C'(y)e^{-y} = xye^{-y}$$

$$C'(y) = xy$$

$$C(x,y) = \frac{1}{2}xy^2$$

$$u = \frac{1}{2}xy^2e^{-y} + C(x)e^{-y}$$

$$u(0,y) = C(0)e^{-y} = 0$$

$$u(x,x) = \frac{1}{2}x^3e^{-x} + C(x)e^{-x} = 0$$

$$\Rightarrow C(x) = -\frac{1}{2}x^3$$

$$u(x,y) = \frac{1}{2}x\,e^{-x}(y^2 - x^2)$$

c)

$u_x + u_y = 0 \qquad u(x,0) = x^2$

$$u(x,y) = \sum_{i,j} a_{ij} x^i y^j$$

$$u(x,0) = \sum_i a_{i0} x^i = x^2 \Rightarrow a_{20} = 1$$

$a_{i0} = 0$ za $i \neq 2$

$$u_x = \sum_{i,j} a_{ij}\, i\, x^{i-1} y^j = a_{10} + \sum_{i,j\geq 0} a_{i+1\,j}\,(i+1) x^i y^j$$

$$u_y = \sum_{i,j} a_{ij}\, j\, x^j y^{j-1} = a_{01} + \sum_{i,j\geq 0} a_{i\,j+1}\,(j+1)\, y^j x^i$$

$$u_x + u_y = a_{01} + a_{10} + \sum_{i,j\geq 1} (a_{i+1,j}(i+1) + a_{i,j+1}(j+1)) x^i y^j = 0$$

$$\Rightarrow (a_{i+1})(i+1) + a_{i,j+1}(j+1) = 0 \text{ za } \forall\, xy$$

$a_{01} + a_{10} = 0 \Rightarrow a_{01} = -a_{10}$

$\Rightarrow a_{0j} = 0$ razen $a_{02} = -2$

$a_{2,1} = a_{1,2}$

$a_{11} = -a_{0,2}$

$a_{11} = -a_{2,0}$

$a_{1,0} + a_{01} = 0,\ a_{1,1} + a_{02} = 0 \Rightarrow a_{11} = 2$

j \ i	0	1	2	3		
0	a_{00}	0	1	0	0	...
1	0	-2	0	0		
2	1	0	0			
3	0	0				
	0					

$a_{3,0} = -a_{2,1}$

$a_{11} = -a_{02}$

$$u(x,y) = a_{00} + a_{2,0} x^2 + a_{11} xy + a_{02} y^2$$

$$= a_{00} + x^2 - xy + y^2$$

$$u(x,0) = a_{00} + x^2 \Rightarrow a_{00} = 0$$

$\Rightarrow u(x,y) = x^2 - xy + y^2 \qquad a_{2,0} = a_{11}$

Hmmm nekaj je narobe

$u(x,y) = x^2 - 2xy + y^2$

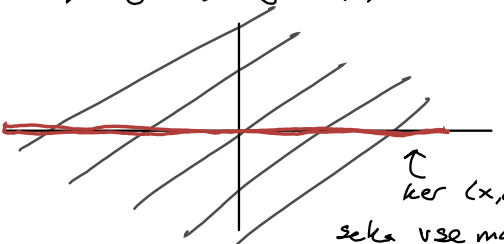
Pozabila sem množiti z $i+1$ in $j+1$ pri odvajanju

3. c)

$u_x + u_y = 0 \qquad u(x,0) = x^2 \quad \rightsquigarrow u(x,y) = (x-y)^2$

Naj bo $u(x,y) = g(x-y)$

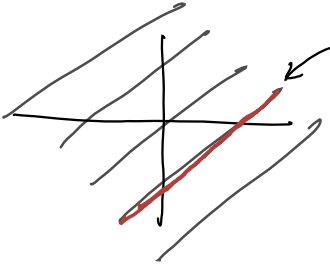
$\Rightarrow u_x + u_y = g'(x-y) - g'(x-y) = 0$



ker $(x,0)$ seka vse možnosti, je to validn način

Slab primer

$u(x,x) = x^2$



če imamo tak robni pogoj ne moremo dobiti rešitve

Ali neobstaja, ali pa jako je veliko

$$u_{xy} + y u_y + u = 1 \qquad u_x(x,0) = 1$$
$$u(x,x) = e^{-x^2}$$

$$y u_y + u = (yu)_y$$

$$u_{xy} + (yu)_y = 1 \rightsquigarrow u_x + yu = y + C(x)$$

$$V(x,0): \quad \overset{0}{\overset{\shortparallel}{u_x(x,0)}} + 0 \cdot 1 = 0 + C(x) \Rightarrow C(x) = 0$$

$$u_x + yu = y$$

homogeno:

$$u_x = -yu$$
$$\frac{\partial u}{u} = -y\,dx$$
$$u = D(y) e^{-yx}$$

partikularna:

$$D_x(x,y) e^{-yx} = y$$
$$\rightsquigarrow D(x,y) = \int y e^{+yx}\,dx = e^{+yx} + C(y)$$

$$u = 1 + C(y) e^{-yx}$$
$$u(x,x) = 1 + C(x) e^{-x^2} = e^{-x^2} \Rightarrow C(x) = \frac{e^{-x^2} - 1}{e^{-x^2}} = 1 - e^{x^2}$$

$$u(x,y) = 1 + (1 - e^{-y^2}) e^{-yx}$$

$$u_{xy} + 2u_y = e^{-2x} \qquad \begin{cases} u(x,0) = e^{-x} \\ u_x(x,0) = -e^{-x} \end{cases} \qquad u(0,y) = 1$$

---

integriamo per y

$$\rightsquigarrow \quad u_x + 2u = ye^{-2x} + C(x)$$

$$v(x,0): \quad -e^{-x} + 2e^{-x} = C(x) = e^{-x}$$

$$u_x + 2u = ye^{-2x} + e^{-x}$$

hom:

$$\frac{\partial u}{u} = -2dx \Rightarrow u = e^{-2x} C(y)$$

part: $\quad C_x(x,y)e^{-2x} = ye^{-2x} + e^{-x}$

$$C(x,y) = \int (y + e^{x})dx = yx + e^{x} + D(y)$$

$$u = yxe^{-2x} + e^{-x} + e^{-2x} D(y)$$

$$v(0,y): \quad 0 + 1 + D(y) = 1 \Rightarrow D(y) = 0$$

$$\Rightarrow u = yxe^{-2x} + e^{-x}$$

KLE

$$a(x,y,u)\,u_x + b(x,y,u)\,u_y = c(x,y,u)$$

$S = \{ (x,y,z) \in \mathbb{R}^3 ;\quad z = u(x,y) \quad u \text{ rešitev} \}$

je integralska krivulja

Presek $S_1 \cap S_2$ je karakteristična krivulja

Karakteristična krivulja γ: $\dot{\gamma}(t) = (a(\gamma(t)), b(\gamma(t)), c(\gamma(t)))$

Izrek:

- S_1, S_2 integralski ploskvi za isto KLE. $p \in S_1 \cap S_2 \Rightarrow$ karakteristična krivulja, ki vsebuje p leži cela v $S_1 \cap S_2$
- $S_1 \cap S_2$. če se ne sekata tangentno, je karakteristična krivulja

Dokaz 2)

$p \in C = S_1 \cap S_2$

$\Pi_1 = T_p S_1 \qquad \Pi_2 = T_p S_2$

$\dim(\Pi_1 \cap \Pi_2) = 1$, ker se ne sekata S_1 in S_2 tangentno.

$\Pi_1 \cap \Pi_2$ tangentna na C

$$a(p), b(p), c(p) \in \Pi_1 \cap \Pi_2 = T_p C$$

$$T_p C = \{\lambda (a(p), b(p), c(p)) + p \quad ; \lambda \in \mathbb{R}\}$$

C je tangentna na karakteristično smer

$\Rightarrow C$ je karak. krivulja

1

$$x u_x + y u_y = 0 \qquad (x, y, 0) = (a, b, c)$$

$$\dot{x} = x \rightsquigarrow x_0(s) e^t$$

$$\dot{y} = y \rightsquigarrow y_0(s) e^t$$

$$\dot{z} = 0 \rightsquigarrow z = z_0(s)$$

$$\varphi(t) = u(tx, ty)$$

$$\dot{\varphi}(t) = u_x \cdot x + u_y \cdot y = 0$$

$$\Rightarrow \varphi(t) = C(s) = C \qquad \varphi(0) = u(0,0)$$

$$\Rightarrow \varphi(t) = u(0,0)$$

$$\varphi(1) = u(x,y) = C = u(0,0)$$

$$\Rightarrow u(x,y) = u(0,0)$$

② $u : \bar{D} \to \mathbb{R}$

$a(x,y)\, u_x + b(x,y)\, u_y = -u$

$x\, a(x,y) + y\, b(x,y) > 0$ za $x,y \in \partial D$

min, max $D \Rightarrow u \equiv 0$

max $\in D \Rightarrow u_x = u_y = 0$ v (x_0, y_0)

$\Rightarrow -u(x_0, y_0) = 0$

min $\in D \Rightarrow -u(x_1, y_1) = 0 \Rightarrow$ $\Rightarrow u \equiv 0$

b) $(x,y,\lambda) \longmapsto u(x,y) + \lambda(x^2+y^2-1)$

$u_x(x,y) + 2\lambda x = 0$

$u_y + 2\lambda y = 0$

$x^2+y^2-1 = 0$

c) $t \xmapsto{f} u(tx_0, ty_0)$ x_0, y_0 maks na robu

$\dot{f} = u_x x_0 + u_y y_0 \geq 0$ v okolici $t=1$

$t=1: u_x(x_0,y_0)\, x_0 + u_y(x_0,y_0)\, y_0 = -2\lambda(x_0 + y_0)$

$\Rightarrow \lambda \leq 0$

Denimo da $u(x_0,y_0) > 0$

$a(x,y)(-2\lambda x_0) + b(x,y)(-2\lambda y_0) \overset{=-u}{<} 0$

$(-2\lambda)\underbrace{(a(x,y)x_0 + b(x,y)y_0)}_{>0} < 0$

$\Rightarrow -2\lambda < 0 \Rightarrow \lambda > 0$ $*$

Sledi $u(x_0,y_0) < 0$

kjer koli je minimum ne v kotu $\geq 0 \Rightarrow$

$u \equiv 0$

Metoda karakteristik za enačbe 1. reda

enačba je

- $a(x, u(x)) \cdot Du = b(x, u(x))$
- $u(x) = g(x)$ .... začetni pogoj
  na krivulji Γ $\quad g : \Gamma \to \mathbb{R}$

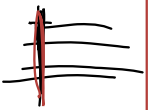
① Rešimo sistem za vsaj po s

$$\dot{z}(y,s) = b(x(y,s), z(y,s))$$

odvod po s

$$\dot{x}(y,s) = a(x(y,s), z(y,s))$$

$$x(y,0) = y \quad ; \quad z(y,0) = g(y)$$



② Najdemo $s(x)$, $y(x)$, potem $u(x) = z(y(x), s(x))$

(1) $u(0,x) = g(x)$

$u_t + a \cdot \nabla u = 0$

$\dot{z}(y,s) = 0 \quad \rightsquigarrow \quad z = c(y)$

$\dot{t}(y,s) = 1 \quad \rightsquigarrow \quad t = 1s + c(y)$

$\dot{x}(y,s) = a \quad \rightsquigarrow \quad x = as + c(y)$

$(t(y,0), x(y,0)) = (0,y)$

$z(y,0) = g(y)$

$z = c(y) = g(y)$

$t(y,0) = c(y) = 0 \rightsquigarrow t(y,s) = s$

$x(y,0) = c(y) = y \rightsquigarrow x(y,s) = as + y$

$y = x - at$

$u(x,y) = g(x - at)$

b)

$$u_t + a\partial u + u = f(x)$$

$$u_t + a\partial u = f(x) - u$$

$$\dot{z}(y,s) = f(x(y,s)) - z(y,s)$$

$$\dot{t}(y,s) = 1 \rightsquigarrow t = s + C(y)$$

$$\dot{x}(y,s) = a \rightsquigarrow x = s + D(y)$$

$$x(y,0) = g(y) \rightsquigarrow x = s + g(y)$$

$$t(y,0) = 0 \rightsquigarrow t = s$$

$$\rightsquigarrow \dot{z}(y,s) = f(s + g(y)) - z$$

hom: $\int \frac{dz}{z} = -\int ds \rightsquigarrow z = C(y)e^{-s}$

par: $\dot{C}(y,s)e^{-s} = f(s + g(y))$

$$\rightsquigarrow C = \int f(s + g(y))\, e^{s})\, ds$$

$$z(y,s) = \int f(t + g(y))\, e^{t}\, dt\; e^{-s}$$

$$= \int f(t)\, e^{t}\, dt\; e^{-g(y)} e^{-s}$$

$$u(x,y) = \left(\int f(t)\, e^{t}\, dt\right) e^{-x}$$

c) $u_t + \nabla F(u) = 0$

$$\nabla F(u(x_1, \dots, x_n)) = \left(F'(u)\frac{\partial u}{\partial x_1}, \dots, F'(u)\frac{\partial u}{\partial x_n} \right) =$$
$$= F'(u) \cdot \nabla u$$

$\dot{z} = 0 \quad \rightsquigarrow \quad z = c(y) = g(y)$

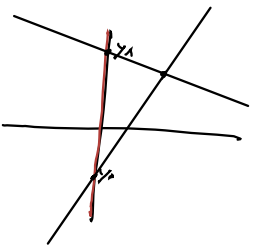
$\dot{x} = F'(z) \rightsquigarrow \dot{x} = F'(c) \rightsquigarrow x = F'(g(y))s + y$

$\dot{t} = 1 \quad \rightsquigarrow \quad t = s + D(y) \rightsquigarrow t = s$

$t(y,0) = 0$

$x(y,0) = y$

$z(y,0) = g(y)$



$x(y,s) = F'(g(y))\, s + y$

kej ko se seceta

$F'(g(y_0))s + y_0 = F'(g(y_1))s + y_0$

če se seceta mora biti $g(y_0) = g(y_1)$

torej če se seceta, se ne more zgoditi
analitična rešitev obstaja brezhibno

a) $uu_x + u_y = 1 \qquad u(x,x) = \frac{x}{2}$

$a = u \qquad \dot{x} = z \rightsquigarrow x = \frac{1}{2}t^2 + \frac{s}{2}t + C(s) \rightsquigarrow$

$b = 1 \qquad \dot{y} = 1 \rightsquigarrow t + C(s) \rightsquigarrow y = t + s$

$c = 1 \qquad \dot{z} = 1 \rightsquigarrow t + C(s) \rightsquigarrow z = t + \frac{s}{2}$

$x(s,0) = s$

$y(s,0) = s$

$z(s,0) = \frac{s}{2}$

$x = \frac{1}{2}t^2 + \frac{1}{2}st + s \rightsquigarrow s = -$

$t = y - s \rightsquigarrow s = y - t$

$u(x,y) = \frac{s}{2} + t = \frac{s}{2} + y - s = y - \frac{s}{2}$

$x = \frac{1}{2}t^2 + \frac{1}{2}yt - t^2 + y - t = -\frac{1}{2}t^2 + \frac{1}{2}yt + y - t$

$\vdots \qquad s = \frac{3x - y^2}{2 - y}$

čo $x \neq 2$ je vrcholu

a z. vrch. $y \neq 2$

b) $-\sin(y)\, u_x + x u_y = 0 \qquad u(x,0) = g(x) \quad za\ x>0$

$$\left.\begin{aligned} \dot{x} &= -\sin(y) \\ \dot{y} &= x \end{aligned}\right\} \qquad \ddot{y} = \dot{x} = -\sin(y)$$

$$\frac{d^2}{dt^2}\sin(y) = \frac{d}{dt}\cos y \cdot \dot{y} = \ddot{y}\cos y + \dot{y}(-\sin y)\dot{y}$$

$$\dot{z} = 0 \rightsquigarrow z = g(s)$$

$$x(s,0) = x$$

$$y(s,0) = 0$$

$$z(s,0) = g(s)$$

1. energija $E_1(x,y,z) = z$

$$\ddot{y} = \dot{x} = -\sin(y) \qquad x\dot{x} = \cancel{x}\ddot{y} = \dot{y}\ddot{y} = -x\sin(y) = -\dot{y}\sin y$$

2. energija

$$x\dot{x} = (\cos y)^{\cdot}$$

$$\int x\,dx \int (\cos y)^{\cdot}\,dt = \cos y + C$$

$$\frac{1}{2}x^2 = \cos y + c$$

$$E_2 = \frac{1}{2}x^2 - \cos y$$

Iščemo H:

$$E_1(x(0), y(0), z(0)) = g(s)$$

$$E_2(x(s,0), y(s,0), z(s,0)) = \frac{1}{2}s^2 - 1$$

$$H(a,b) = a - g(\sqrt{2b+2})$$

$$H(E_1, E_2) = g(s) - g(\sqrt{s^2}) = 0$$

$$H(E_1(x,y,z), E_2(x,y,z)) = 0 \quad za\ vse\ t,s$$

$$\overset{\shortparallel}{z}$$

$$z - g(\sqrt{x^2 - 2\cos y + 2}) = 0$$

$$\Rightarrow u(x,y) = g(\sqrt{x^2 - 2\cos y + 2})$$

c) $x(y^2+u)u_x - y(x^2+u)u_y = (x^2-y^2)u$

$u(x,x) = c \in \mathbb{R}$ za $x \neq 0$

$\dot{x} = x(y^2+z)$

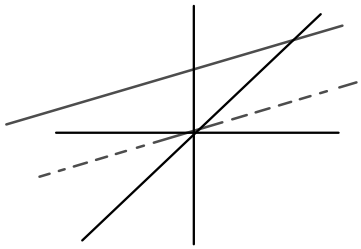
$\dot{y} = -y(x^2+z)$

$\dot{z} = z(x^2-y^2) = z(x-y)(x+y)$

$x(s,0) = s$

$y(s,0) = s$

$z(s,0) = c$



$\dot{x}(s,0) = s(s^2+c)$

$\dot{x}x + \dot{y}y = z(x^2-y^2) = \dot{z}$

$\frac{x^2}{2} + \frac{y^2}{2} = z + D(s)$

$E_1(x,y,z) = \frac{x^2}{2} + \frac{y^2}{2} - z$

$\frac{\dot{x}}{x} + \frac{\dot{y}}{y} + \frac{\dot{z}}{z} = y^2 - x^2 + x^2 - y^2 = 0$

$\ln x + \ln y + \ln z = c$

$xyz = F(s) \qquad E_2(x,y,z) = xyz$

$E_1(s,0) = s^2 - c = a$

$E_2(s,0) = cs^2 = b \qquad ca - b + c^2$

$H(a,b) = ca - b + c^2$

$H(E_1,E_2) =$

$c\left(\frac{x^2}{2} + \frac{y^2}{2} - z\right) - xyz + c^2 = 0$

izrazimo z

$$z = u(x,y) = \frac{c^2 + \frac{c}{2}(x^2+y^2)}{c + xy}$$

Energija

Pri uporabi karakteristik v n-dimenzionalnem sistemu, lahko iščemo n "energij". To so funkcije $E_j(x(t))$ za katere velja, da so

$$\frac{d}{dt} E_j(x(t)) = 0$$

Nato predpišemo $H(E_1, \ldots, E_n) = 0$ (oz poljubno konstanto) in z uporabo začetnih pogojev določimo H, ki zadošča $H(E_1(x(0)), \ldots) = 0$